

RPM

RedHat Package Manager

Package Naming Convention

`packagename-a.b.c-x.arch.rpm`

<code>packagename</code>	name of package
<code>a.b.c</code>	version number
<code>x</code>	build/release minor changes to build
<code>arch</code>	cpu architecture e.g. i386 for any x86 CPU from 80386 onwards <code>noarch</code> cpu-independent package <code>src</code> source package

Compatibility issues

packages can use linux distro files that are not present on your distro
e.g. a package may require a newer version of a library than you have

Commands

package and files

- filename may be `samba-4.1.9-4.fc20.x86_64.rpm`
- package name is `samba`
- install multiple packages by separating with spaces

`rpm [operation][options] [package/file]`

operations

<code>-i</code>	install
<code>-U</code>	install or upgrade
<code>-F --freshen</code>	upgrade
<code>-e</code>	uninstall
<code>-q</code>	query - checked if installed, files it contains etc
<code>-V --verify</code>	verify - check files unchanged since install
<code>-b</code>	build package - from source code and config files

options

<code>-h --hash</code>	<code>-i, -U, -F</code>	display # to show progress
<code>--nodeps</code>	<code>-i, -u, -F, -e</code>	no dependency checks be performed
<code>--test</code>	<code>-i, -U, -F</code>	check for dependency conflicts without installing package
<code>--prefix path</code>	<code>-i, -U, -F</code>	set installation directory
<code>-i</code>	<code>-q</code>	display package info

examples

install or upgrade a package

```
rpm -Uh samba-4.1.8-4.fc20.x86_64.rpm
```

-U install or upgrade

-h display # progress bar

display package info

```
rpm -qi sambs
```

-q check if installed

-i display info

Yum

alternative to `rpm`

automatically locates packages by searching its repositories

```
yum [options] [command] [package]
```

command	
<code>install</code>	
<code>update</code>	update or
<code>check-update</code>	check if updates available
<code>remove</code> <code>erase</code>	uninstall
<code>list</code>	display package info

options available depend on the command# Extracting data from RPMs

Extra data from RPM without installing the package

e.g. to view the original source code, fonts etc

Steps

- convert to cpio file

use `rpm2cpio` program

```
rpm2cpio package.rpm > package.cpio
```

- extract the data

```
cpio -i --make-directories < package.cpio
```

-i extract

-make-directories creates directories

Configuring RPM/Yum package managers

The main use is to add architecture optimizations for your CPU

RPM

the main config file `/usr/lib/rpm/rpmrc` - you shouldn't edit this file
instead create and edit `/etc/rpmrc`

example optimizations: `optflags: athlon -O2 -g -march=i686`

this line tells RPM when building for athlon platform use those options

Yum

configured via:

`/etc/yum.conf`

`/etc/yum.repos.d/` - a directory, each has files that describe a repository